



Material simulation

Finding certifiable bounds on ground-state properties beyond
energy for Silicon

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Introduction

Finding the ground state energy of interacting many-body quantum systems is often considered the ‘holy grail’ of computational physics, as it is a quantity of central importance both for fundamental theory and for applications (materials, chemistry, nuclear physics, etc.). However finding an exact solution to this problem is not doable as the dimensions for the Hilbert spaces grow exponentially with the size of the system, and so far exact solutions are only known for simplified systems such as the one-dimensional Heisenberg model. Obviously many systems physicists are interested in are much more complex such as a new material for battery cathodes or solar panels.

Given this situation many methods for approximating the ground state energies have been fashioned over the last decades, a class of which are variational methods [1] that aim to finding the minimum of the expected value of the Hamiltonian which always lies above the ground state energy of the system E_0 by definition.

While variational methods, in this sense, are not new, a particularly promising development in this area is the Variational Quantum Eigen-solver (VQE). VQE leverages the special capabilities of quantum processors to represent trial wavefunctions efficiently, offering the prospect of tackling problems that are intractable for classical computers. Indeed, VQE is often cited as one of the strongest near-term motivations for building universal quantum computers [2].

In the context of this new method comes an interesting paper by Phasecraft [3] published in Nature Communications, and for which a much more detailed version is given on their website entitled "Towards near-term quantum simulations of materials" demonstrating that it is possible to reduce the requirements for running a functional VQE on a near-term quantum processor. By reducing the depth of the quantum circuits required to run it they relax the requirements on the coherence of the qubits, as they also reduce the number of qubits required.

This paper is the main inspiration for this work. We planned to use the Hamiltonian constructed by Phasecraft in their study as a testbed for evaluating a family of classical algorithms for the ground energy problem namely ones based on *Semi-Definite Programming*. Importantly, such methods give rigorous lower bounds on the ground state energy. Semi-definite programs (SDP) are a class of convex optimization programs, and as the name suggests it uses semi-definite matrices to describe the problem. The idea is that it tries to minimize the expectation value while imposing constraints that keep a certain matrix semi-definite and satisfying other constraints.

Like Variational methods SDP’s are not particularly new [4], the mathematical tools used in it were under developments since the 1970’s and crystallized as a separate area in the 1990’s and was used since in a wide variety of fields and became a standard tool for quantum information theorists. Its use in many-body physics problems did not become as popular until recently [5] used as a tool for what is known as *bootstrapping quantum systems*. Thus, the literature on simulating many-body quantum systems using SDP’s remains relatively limited, though interest has been increasing in recent years, which makes it an interesting area to study.

We will also see later (2) that SDP’s are particularly beneficial as they cover for shortcomings of variational methods and can actually be used in combination to find interesting results[6].

This being said, the remainder of this report is organized as follows. Chapter 1 presents

the formalism and theoretical background used in the Phasecraft paper to construct the Hamiltonian. It also describes the workflow required to obtain it in practice, and a Github repository has been created to include all the scripts used in the Silicon example. Chapter 2 presents an introduction to semi-definite programming and its use for quantum bootstrapping and how they can be used to certify ground state properties of many-body systems. Finally, in chapter 3 we attempt to present an implementation of SDP's as an alternative for VQE's used in the Phasecraft paper, in order to find a lower bound for silicon—a semiconductor that is heavily studied due to its extensive use in electronic industries.

Chapter 1

Effective description of the Hamiltonian:

In this Chapter we present the formalism used in the Phasecraft study [3], keeping the same notations and the same workflow with certain adjustments to make it suitable for this work, while also giving a more elementary explanation and more details for certain stages of the Hamiltonian construction.

1.1 General characteristics of Fermion Hamiltonian

For constructing the Hamiltonian of a crystal (Silicon in the context of this work) to take into consideration the electron-electron interaction which is usually omitted or approximated while finding ground state energy, we use the second-quantization formalism with creation and annihilation operators.

Thus, the most general form of the Hamiltonian for the purposes of this work is :

$$\begin{aligned}\hat{H} = & \sum_{\sigma} \int d^3\vec{r} \, \hat{\psi}_{\sigma}^{\dagger}(\vec{r}) \left[-\frac{\hbar^2}{2m} \nabla^2 + U(\vec{r}) \right] \hat{\psi}_{\sigma}(\vec{r}) \\ & + \frac{1}{2} \sum_{\sigma, \sigma'} \int d^3\vec{r} d^3\vec{r}' \, \hat{\psi}_{\sigma}^{\dagger}(\vec{r}) \hat{\psi}_{\sigma'}^{\dagger}(\vec{r}') V(\vec{r} - \vec{r}') \hat{\psi}_{\sigma'}(\vec{r}') \hat{\psi}_{\sigma}(\vec{r})\end{aligned}\tag{1.1}$$

Notation:

- $\hat{\psi}^{\dagger}$ ($\hat{\psi}$) : Field creation (annihilation) operators.
- $V(\vec{r} - \vec{r}')$: Distance-dependent repulsive potential between electrons.

We use the screened Coulomb potential:

$$V(\vec{r} - \vec{r}') = \frac{q_e^2}{4\pi\epsilon_0} \frac{e^{-\mu|\vec{r}-\vec{r}'|}}{|\vec{r} - \vec{r}'|},\tag{1.2}$$

where μ is the inverse screening length.

External potential created by the ions in the lattice: In the spirit of the Born-Oppenheimer approximation, ions are frozen, and they create a periodic external potential expressed as :

$$U(\vec{r}) = \frac{q_e}{4\pi\epsilon_0} \sum_{\vec{R}_i} \frac{Ze}{|\vec{r} - \vec{R}_i|},\tag{1.3}$$

with Ze the charge of the ion and the sum taken over lattice sites $\vec{R}_i$. $U(\vec{r})$ is periodic.

1.1.1 Structure of two and four Fermion integrals

We expand the field operator in a wavefunction basis $\{\phi_\lambda(\vec{r})\}$:

$$\hat{\psi}_\sigma(\vec{r}) = \sum_\lambda \phi_\lambda(\vec{r}) c_{\lambda,\sigma}, \quad (1.4)$$

where $c_{\lambda,\sigma}$ are annihilation operators in state (λ, σ) , where λ is the set of quantum numbers describing each wavefunction and σ is the spin.

Substituting and simplifying, the Hamiltonian becomes:

$$\hat{H} = \sum_{\lambda_1, \lambda_2, \sigma} t_{\lambda_1, \lambda_2} c_{\lambda_1, \sigma}^\dagger c_{\lambda_2, \sigma} + \frac{1}{2} \sum_{\lambda_1, \lambda_2, \lambda_3, \lambda_4} V_{\lambda_1, \lambda_2, \lambda_3, \lambda_4} c_{\lambda_1, \sigma}^\dagger c_{\lambda_2, \sigma'}^\dagger c_{\lambda_3, \sigma'} c_{\lambda_4, \sigma}, \quad (1.5)$$

with the usual fermionic anti-commutation rules $c_{\lambda, \sigma} c_{\lambda', \sigma'}^\dagger + c_{\lambda', \sigma'}^\dagger c_{\lambda, \sigma} = \delta_{\lambda \lambda'} \delta_{\sigma \sigma'}$

The Hopping Matrix (HM) and the Coulomb Tensor (CT) elements are defined respectively as :

$$t_{\lambda_1, \lambda_2} = \int d^3 \vec{r} \phi_{\lambda_1}^*(\vec{r}) \left[-\frac{\hbar^2}{2m} \nabla^2 + U(\vec{r}) \right] \phi_{\lambda_2}(\vec{r}). \quad (1.6)$$

$$V_{\lambda_1 \lambda_2 \lambda_3 \lambda_4} = \frac{1}{2} \int d^3 \vec{r} \int d^3 \vec{r}' \phi_{\lambda_1}^*(\vec{r}) \phi_{\lambda_2}^*(\vec{r}') V(\vec{r} - \vec{r}') \phi_{\lambda_3}(\vec{r}') \phi_{\lambda_4}(\vec{r}) \quad (1.7)$$

We observe CT obeys index-swap symmetry : $V_{\lambda_1 \lambda_2 \lambda_3 \lambda_4} = V_{\lambda_2 \lambda_1 \lambda_4 \lambda_3}$

While HM is Hermitian : $t_{\lambda_1, \lambda_2} = t_{\lambda_2, \lambda_1}^*$.

Cauchy–Schwarz inequality for the CT

We introduce the notation : $\rho_{\lambda_i \lambda_j}(\vec{r}, \vec{r}') = \phi_{\lambda_i}^*(\vec{r}) \phi_{\lambda_j}(\vec{r}')$. This allows us to exploit the CT expression as the usual inner product for functions:

$$\langle \rho_{\lambda_1 \lambda_4}, \rho_{\lambda_2 \lambda_3} \rangle_1 = V_{\lambda_1 \lambda_2 \lambda_3 \lambda_4}$$

Applying the Cauchy–Schwarz inequality gives:

$$|V_{\lambda_1 \lambda_2 \lambda_3 \lambda_4}|^2 \leq V_{\lambda_1 \lambda_4 \lambda_1 \lambda_4} V_{\lambda_2 \lambda_3 \lambda_2 \lambda_3} \quad (1.8)$$

Which would come in handy to reduce the needed variables in simulating later.

1.2 Real-space single-particle basis

1.2.1 Beyond Bloch States

In solid-state physics, the electronic structure of a periodic crystal is described by Bloch states, $|\psi_{n\mathbf{k}, \sigma}\rangle$, which are eigenstates of the Hamiltonian and are delocalized over the entire crystal. These states are characterized by a crystal momentum $\mathbf{k}$ (within the Brillouin zone), a band index n , and a spin σ . Their wavefunction has the form

$$\psi_{n\mathbf{k}, \sigma}(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} u_{n\mathbf{k}, \sigma}(\mathbf{r})$$

where $u_{n\mathbf{k}, \sigma}(\mathbf{r})$ is a function with the same periodicity as the crystal lattice. While this basis is natural for calculating band structures, a real-space, localized basis is often desirable for interpreting chemical bonding, computing specific properties like polarization,

and building model Hamiltonians.

An alternative to the extended Bloch orbital representation is the Wannier function representation, originally introduced by Gregory Wannier in 1937 [7]. Marzari et al. (2012) [8] developed a systematic method for constructing a unique set of *Maximally localized Wannier functions* (MLWFs), providing a powerful tool for electronic structure analysis and applications in condensed matter physics. Unlike Bloch orbitals, which are delocalized across the entire crystal, MLWFs exhibit spatial localization as their amplitude vanishes quickly away from their centers while respecting crystal symmetries. This localization enables the use of a minimal and physically intuitive basis set for describing electronic properties, simplifying calculations (see Figure 1.1 from [8]).

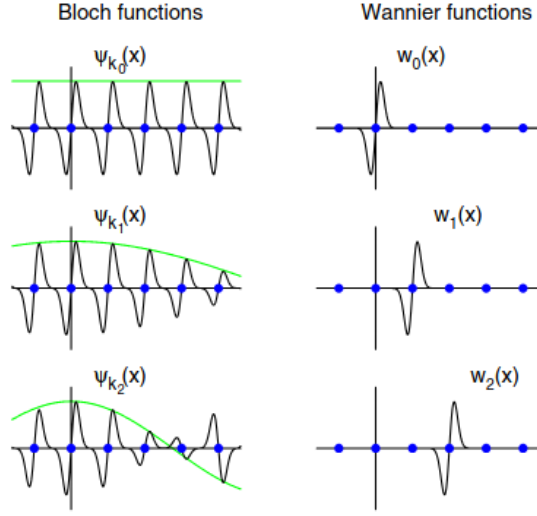


Figure 1.1: Comparison between Bloch functions and Wannier functions (WFs). Left: Real-space representation of three of the Bloch functions $e^{i\mathbf{k}\cdot\mathbf{r}}u_{n\mathbf{k},\sigma}(\mathbf{r})$ associated with a single band in 1D, for three different values of the wave vector $\mathbf{k}$. Filled circles indicate lattice vectors, and thin lines indicate the $e^{i\mathbf{k}\cdot\mathbf{r}}$ envelopes of each Bloch function. Right: WF associated with the same band, forming periodic images of one another. The two sets of Bloch functions at every $\mathbf{k}$ in the Brillouin zone and WFs at every lattice vector span the same Hilbert space.

1.2.2 Definition and Gauge Freedom

The Wannier function for band n centered at lattice vector $\mathbf{R}$ is formally defined as the Fourier transform of the Bloch states over the Brillouin zone (BZ):

$$|w_{n\mathbf{R}}\rangle = \frac{V}{(2\pi)^3} \int_{\text{BZ}} e^{-i\mathbf{k}\cdot\mathbf{R}} |\psi_{n\mathbf{k}}\rangle d\mathbf{k} \quad (1.9)$$

where V is the unit cell volume. These functions form a complete, orthonormal set $\langle w_{m\mathbf{R}} | w_{n\mathbf{S}} \rangle = \delta_{mn} \delta_{\mathbf{RS}}$, and are typically localized in real space.

A critical aspect of this definition is its inherent *gauge freedom*. The Bloch states are not unique; any $\mathbf{k}$ -dependent phase transformation

$$|\tilde{\psi}_{n\mathbf{k}}\rangle = e^{i\phi_n(\mathbf{k})} |\psi_{n\mathbf{k}}\rangle \quad (1.10)$$

yields equally valid eigenstates. This phase choice $\phi_n(\mathbf{k})$ profoundly affects the spatial localization of the resulting Wannier function defined via Eq. 1.9. A naive, arbitrary choice of gauge typically results in WFs that are poorly localized and spread over many unit cells, limiting their practical utility.

For a group of J bands, this freedom generalizes to a unitary mixing of the Bloch states at each $\mathbf{k}$ -point:

$$|\tilde{\psi}_{\mathbf{k}}\rangle = \sum_{m=1}^J U_{mn}^{(\mathbf{k})} |\psi_{m\mathbf{k}}\rangle, \quad (1.11)$$

where $\mathbf{U}^{(\mathbf{k})}$ is a $J \times J$ unitary matrix for each $\mathbf{k}$ -point ($\mathbf{U}^{(\mathbf{k})}\mathbf{U}^{(\mathbf{k})\dagger} = \mathbf{I}$). The generalized Wannier function is then:

$$|w_{n\mathbf{R}}\rangle = \frac{V}{(2\pi)^3} \int_{\text{BZ}} e^{-i\mathbf{k}\cdot\mathbf{R}} \left(\sum_{m=1}^J U_{mn}^{(\mathbf{k})} |\psi_{m\mathbf{k}}\rangle \right) d\mathbf{k}. \quad (1.12)$$

The set of matrices $\mathbf{U}^{(\mathbf{k})}$ for all $\mathbf{k}$ -points constitutes the gauge choice and entirely determines the properties of the Wannier functions. Note that if the matrix $\mathbf{U}^{\mathbf{k}}$ which is easiest gauge but not the optimal, then the wannier states are the usual Fourier transform of the Bloch states and the hamiltonian is just using creation and annihilation operators indexed by the atomic site and quantum atomic numbers as would be the case in the example used for Graphene under tight binding model in Section 2.2.

1.2.3 New Expression of the Hamiltonian in the Wannier Basis

In the Wannier basis, the Hamiltonian from Eq. 1.5 can be expressed as:

$$\begin{aligned} H^W = & \sum_{\sigma} \sum_{\substack{m,n \\ \mathbf{R}_1, \mathbf{R}_2}} T(\mathbf{R}_1 - \mathbf{R}_2)_{mn} w_{\mathbf{R}_1, m, \sigma}^{\dagger} w_{\mathbf{R}_2, n, \sigma} \\ & + \sum_{\sigma, \sigma'} \sum_{\substack{s, l, m, n \\ \mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3, \mathbf{R}_4}} \tilde{V}_{slmn}^{(\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3, \mathbf{R}_4)} w_{\mathbf{R}_1, s, \sigma}^{\dagger} w_{\mathbf{R}_2, l, \sigma'}^{\dagger} w_{\mathbf{R}_3, m, \sigma'} w_{\mathbf{R}_4, n, \sigma}, \end{aligned} \quad (1.13)$$

where the operators $w_{\mathbf{R}, m, \sigma}^{\dagger}$ create electrons in Wannier states localized near site $\mathbf{R}$ with orbital index m and spin σ .

The Wannier functions associated with these operators are constructed from Bloch wavefunctions as:

$$\mathcal{W}_{s, \sigma}^{\mathbf{R}}(\mathbf{r}) = \mathcal{W}_{s, \sigma}^0(\mathbf{r} - \mathbf{R}) = \sum_{\mathbf{k}, n} e^{-i\mathbf{k}\cdot\mathbf{R}} U_{ns}(\mathbf{k}) u_{\mathbf{k}, n, \sigma}(\mathbf{r}) e^{i\mathbf{k}\cdot\mathbf{r}}, \quad (1.14)$$

where $U(\mathbf{k})$ are unitary matrices defining the Wannier gauge.

With these, the hopping matrix elements and Coulomb interaction tensor can be expressed as:

$$T(\mathbf{R}_1 - \mathbf{R}_2)_{mn} = \int d\mathbf{r} \mathcal{W}_{m, \sigma}^{\mathbf{R}_1*}(\mathbf{r}) \left[-\frac{\hbar^2 \nabla^2}{2m} + \tilde{U}(\mathbf{r}) \right] \mathcal{W}_{n, \sigma}^{\mathbf{R}_2}(\mathbf{r}), \quad (1.15)$$

$$\tilde{V}_{slmn}^{\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3, \mathbf{R}_4} = \frac{1}{2} \int d\mathbf{r} d\mathbf{r}' \mathcal{W}_{s, \sigma}^{\mathbf{R}_1*}(\mathbf{r}) \mathcal{W}_{l, \sigma'}^{\mathbf{R}_2*}(\mathbf{r}') \mathcal{V}(|\mathbf{r} - \mathbf{r}'|) \mathcal{W}_{m, \sigma'}^{\mathbf{R}_3}(\mathbf{r}') \mathcal{W}_{n, \sigma}^{\mathbf{R}_4}(\mathbf{r}), \quad (1.16)$$

Exploiting the discrete translational invariance of the lattice, these matrix elements can be rewritten relative to the origin as:

$$T(\mathbf{R})_{mn} = \int d\mathbf{r} \mathcal{W}_{m,\sigma}^{\mathbf{R}*}(\mathbf{r}) \left[-\frac{\hbar^2 \nabla^2}{2m} + \tilde{U}(\mathbf{r}) \right] \mathcal{W}_{n,\sigma}^{\mathbf{0}}(\mathbf{r}), \quad (1.17)$$

$$\tilde{V}_{slmn}^{\mathbf{R}_2, \mathbf{R}_3, \mathbf{R}_4} = \frac{1}{2} \int d\mathbf{r} d\mathbf{r}' \mathcal{W}_{s,\sigma}^{\mathbf{0}*}(\mathbf{r}) \mathcal{W}_{l,\sigma'}^{\mathbf{R}_2*}(\mathbf{r}') \mathcal{V}(|\mathbf{r} - \mathbf{r}'|) \mathcal{W}_{m,\sigma'}^{\mathbf{R}_3}(\mathbf{r}') \mathcal{W}_{n,\sigma}^{\mathbf{R}_4}(\mathbf{r}). \quad (1.18)$$

Thanks to the MLWFs, these integrals can be truncated beyond a certain neighbor order—determined by the material properties and required accuracy—thereby substantially reducing computational cost. Furthermore, using lattice symmetries, one only needs to compute integrals around the origin site and subsequently translate the results across the lattice, as reflected in Eqs. 1.18.

1.2.4 Maximally Localized Wannier Functions (MLWFs)

The existence of these MLWFs is guaranteed under two key conditions:

1. The system has a vanishing Chern number. This condition is automatically satisfied in systems with time-reversal symmetry, such as crystals, for which the Chern number is zero. While sufficient, this condition is not strictly necessary, allowing us to proceed without introducing Chern numbers in detail.
2. An energy gap separates the bands in the active space (i.e., the selected bands relevant for the context of this work) from the rest of the spectrum. Note that satisfying this condition does not necessarily imply the system is an insulator, as the Fermi energy may lie within the active space isolated from other bands.

To obtain a unique, optimal, and chemically meaningful set of Wannier functions (WFs), the gauge freedom associated with unitary mixing of Bloch states must be fixed. The method of MLWFs achieves this by selecting unitary matrices $\mathbf{U}^{(\mathbf{k})}$ that minimize the total spatial spread of the Wannier functions, expressed by the spread functional Ω for a set of WFs $\{|w_{n\mathbf{0}}\rangle\}$ as the sum of their second central moments:

$$\Omega = \sum_n \left[\langle w_{n\mathbf{0}} | r^2 | w_{n\mathbf{0}} \rangle - |\langle w_{n\mathbf{0}} | \mathbf{r} | w_{n\mathbf{0}} \rangle|^2 \right], \quad (1.19)$$

where $|\langle w_{n\mathbf{0}} | \mathbf{r} | w_{n\mathbf{0}} \rangle|$ denotes the Wannier center. Minimizing Ω with respect to all $\mathbf{U}^{(\mathbf{k})}$ matrices yields the unique MLWFs.

The resulting MLWFs are typically highly localized, real-valued, and transform according to the point group symmetry of their centers; that is, an MLWF centered on a site inherits the site's point group symmetry. They often resemble atomic or hybrid orbitals (e.g., sp^3), providing a direct bridge between electronic structure calculations and chemical intuition.

This section highlights the advantage of employing the Wannier basis, as it significantly decreases the computational resources necessary for simulating complex materials. In the next one we discuss how we get them in practice. In particular the minimization of the spread defined in Eq. 1.19 which is a complex numerical procedure efficiently handled by the widely used open-source software WANNIER90 [9], the standard tool for this purpose. We shall discuss the computational workflow and the physical principles behind finding MLWFs and implementing them for the construction of Hamiltonian following the same road-map given in the Phascraft [3] study.

1.3 From Electrons to Models: The DFT and Wannier90 Workflow

1.3.1 Density Functional Theory Framework

The computational workflow begins with Density Functional Theory (DFT), a first-principles approach that revolutionized materials modeling by reformulating the complex many-body quantum mechanical problem using the electron density $n(\mathbf{r})$ as the fundamental variable rather than the many-electron wavefunction. This reformulation is based on the Hohenberg-Kohn theorems [10, 11], which establish that all ground state properties of a system are unique functionals of the electron density.

The practical implementation of DFT is achieved through the Kohn-Sham equations [11], which map the interacting many-electron system onto an auxiliary system of non-interacting electrons moving in an effective potential:

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + \bar{U}_{\text{eff}}(\mathbf{r}) \right] \phi_i(\mathbf{r}) = \varepsilon_i \phi_i(\mathbf{r}),$$

where the effective potential $\bar{U}_{\text{eff}}(\mathbf{r})$ contains several components:

$$\bar{U}_{\text{eff}}(\mathbf{r}) = U_{\text{ext}}(\mathbf{r}) + V_{\text{H}}[n(\mathbf{r})] + V_{\text{XC}}[n(\mathbf{r})].$$

Here, $U_{\text{ext}}(\mathbf{r})$ represents the external potential from atomic nuclei, V_{H} is the Hartree potential describing classical electron-electron repulsion, and V_{XC} is the exchange-correlation potential that captures quantum mechanical effects and presents the primary challenge of implementing DFT since it must be approximated [12].

To efficiently handle the rapidly oscillating wavefunctions near atomic cores, DFT calculations employ *pseudopotentials* [13, 14] that replace the strong Coulomb potential of the nucleus and core electrons with an effective potential that reproduces the behavior of valence electrons. This approximation significantly reduces computational cost while maintaining accuracy for chemical bonding and electronic properties.

The Kohn-Sham Hamiltonian in DFT takes the form:

$$\hat{H}_{\text{KS}} = -\frac{\hbar^2}{2m} \nabla^2 + \bar{U}_{\text{eff}}(\mathbf{r}).$$

Diagonalization of this Hamiltonian yields the Kohn-Sham orbitals $\phi_i(\mathbf{r})$ and eigenvalues ε_i that are used to construct the electron density:

$$n(\mathbf{r}) = \sum_i^{\text{occ}} |\phi_i(\mathbf{r})|^2.$$

The self-consistent solution of these equations (represented by the inner loop in Figure 1.2) provides the ground state electron density and energy, which serve as the foundation for subsequent analysis using Wannier functions.

This section outlines the computational pipeline (the same as used by Phasescraft) described in Figure 1.2 for calculating the electronic properties of a material and transforming them into a simplified model. The process involves two main software packages: QUANTUM ESPRESSO, which performs the first-principles calculations, and WANNIER90, which constructs localized orbitals.

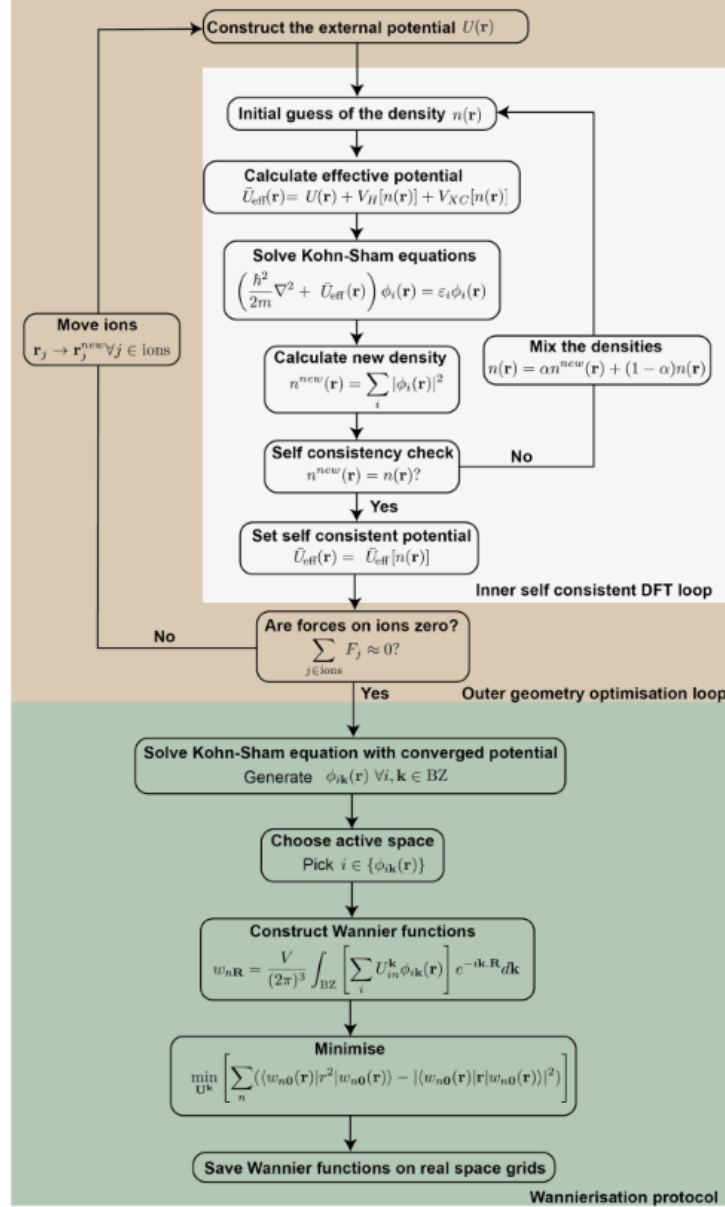


Figure 1.2: The self-consistent workflow for calculating the electronic ground state of a crystal (inner loop), optimizing its geometry (outer loop), and subsequently constructing Maximally Localized Wannier Functions (MLWFs).

1.3.2 The Two-Loop First-Principles Calculation

These loops are handled by QUANTUM-ESPRESSO [15]. It begins with Density Functional Theory, which calculates the total energy and electron distribution (density, $n(\mathbf{r})$) of a system of atoms.

1. **Inner Self-Consistent Loop:** The core task is to solve the Kohn-Sham equations to find the quantum states (wavefunctions $\phi_i(\mathbf{r})$) of electrons in the effective potential of the atomic nuclei and other electrons. This is an iterative process:

- An initial guess is made for the electron density $n(\mathbf{r})$.
- An effective potential U_{eff} is constructed from this density.

- The Kohn-Sham equations are solved with this potential to find new wavefunctions and a new electron density.
- The new density is mixed with the old one, and the process repeats until the density no longer changes (**self-consistency check**), indicating a converged electronic ground state.

2. **Outer Geometry Optimization Loop:** The converged electron distribution is used to calculate forces acting on each atom. If these forces are non-zero, the atomic positions are adjusted, and the entire self-consistent cycle is repeated. This continues until the atoms reach their equilibrium positions.

The final output of this stage is the accurate, self-consistent potential U_{eff} and the corresponding Bloch wavefunctions $\phi_{i\mathbf{k}}(\mathbf{r})$ for all electrons at all $\mathbf{k}$ -points in the Brillouin zone.

1.3.3 Constructing Localized Orbitals

The Bloch wavefunctions from DFT are ideally suited for describing electrons in a perfect crystal but are delocalized over the entire structure. The WANNIER90 [16] code performs a mathematical transformation to create a new, more chemically intuitive basis set:

1. An **active space** of Bloch states (e.g., those forming the valence bands) is selected.
2. A unitary transformation $U_{mn}^{(\mathbf{k})}$ is applied to these states. This transformation is chosen specifically to minimize the spatial spread of the resulting functions, creating **Maximally Localized Wannier Functions (MLWFs)** $w_n(\mathbf{r})$. These functions are localized on atomic sites or bonds, resembling traditional atomic orbitals but derived directly from the quantum-mechanical solution.
3. The matrix elements of the Hamiltonian in this new Wannier basis, $H_{ij} = \langle w_i | \hat{H} | w_j \rangle$, define an effective tight-binding model. These elements represent *hopping integrals* between orbitals.

1.3.4 The Purpose: From ab initio Data to Rigorous Bounds

One may question the need to transform the already-computed DFT solution, since it does already give an approximation of the ground state energy. The power of this workflow lies in the **transferability** and **utility** of the resulting MLWF-based Hamiltonian $\mathcal{H}^{\text{MLWF}}$.

While DFT provides a superb ground-state energy for the specific calculated crystal, the MLWF Hamiltonian is a compact, real-space representation of the material’s electronic structure. It can be used for extremely efficient calculations of properties for very large systems (e.g., defects, surfaces) that are beyond the reach of direct DFT.

Crucially for this project, this first-principles-derived Hamiltonian serves as the input for a more advanced mathematical technique called **Semidefinite Programming (SDP)**. The SDP method can use this Hamiltonian to calculate a mathematically rigorous *lower bound* for the true ground state energy of the system. This serves as a powerful independent check on the DFT result. While DFT provides an approximate energy from one method, the SDP lower bound provides a strict mathematical guarantee: the true quantum mechanical energy of the system *must be* at least this value. This

cross-validation between different computational paradigms strengthens the credibility of the final result.[\[6\]](#).

In the following Chapter, we give a brief overview for SDP methods, how it can be implemented for quantum many-body systems and see a simple example for a small Graphene Lattice.

Chapter 2

Certifying Ground-State Properties of Many-Body with Semi-definite programs

2.1 Overview of Semi-Definite Programs and Their Use in Quantum Bootstrapping

Among quantum many-body problems, the study of ground states plays a central role due to its fundamental relevance for understanding low-energy phases of quantum systems and for characterizing genuine quantum correlations that lack classical analogs. Formally, the ground state $|\psi_{\text{GS}}\rangle$ of a quantum Hamiltonian H is defined as the state of minimal energy, that is a solution to the variational problem

$$E_{\text{GS}} = \min_{|\psi\rangle \in \mathcal{H}} \langle \psi | H | \psi \rangle ,$$

where $\mathcal{H}$ is the Hilbert space of the system, typically $\mathbb{C}^d$ for finite-dimensional systems. The ground-state energy E_{GS} represents the global minimum of the energy functional, and the corresponding quantum state $|\psi_{\text{GS}}\rangle$ encodes key physical properties such as correlations, entanglement, and response to external perturbations.

Due to the exponential growth of $\dim(\mathcal{H})$ with the number of particles—the so-called curse of dimensionality—exact diagonalization of H becomes computationally infeasible for large systems. Accordingly, approximate numerical methods such as variational ansatz are often employed to estimate E_{GS} [1]. Generally, this is done by limiting the minimization problem to a subset of ansatz states $\mathcal{A}$ in the hope that the state $|\psi_{\mathcal{A}}\rangle$, for which the energy is

$$E_{\mathcal{A}} = \langle \psi_{\mathcal{A}} | H | \psi_{\mathcal{A}} \rangle = \min_{|\psi\rangle \in \mathcal{A}} \langle \psi | H | \psi \rangle ,$$

gives results sufficiently close to the real ones; i.e., $E_{\mathcal{A}} \sim E_{\text{GS}}$ and $|\psi_{\mathcal{A}}\rangle \sim |\psi_{\text{GS}}\rangle$, and so are other physically relevant quantities $o_{\mathcal{A}} \sim o_{\text{GS}} = \langle \psi_{\text{GS}} | O | \psi_{\text{GS}} \rangle$.

This approach suffers, however, from two major drawbacks. For starters, there is no mathematical guarantee—although some physical arguments can be given [17]—that $E_{\mathcal{A}} \sim E_{\text{GS}}$, since $E_{\mathcal{A}}$ can correspond to a local minimum and the choice of the subset $\mathcal{A}$ may be inappropriate. More importantly, even if the energies are close, there is no guarantee that the states satisfy $|\psi_{\mathcal{A}}\rangle \sim |\psi_{\text{GS}}\rangle$, therefore giving no trusted information on other ground-state properties.

2.1.1 Relaxations of the Ground-State Problem

A possible way to complement the variational approach is the relaxations of the ground-state problem. While the Ansatz in variational method approaches the hilbert space from the inside as $\mathcal{A} \subset \mathcal{H}$, the relaxations approach a set of possible tuples of expected energies $\mathcal{M}_Q$ from the outside by taking a superset $\mathcal{M} \supset \mathcal{M}_Q$.

More precisely, we take $H = \sum_{i=1}^n \alpha_i h_i \implies E_{\text{GS}} = \min_{\{\langle h_i \rangle\} \in \mathcal{M}_Q} \sum_{i=1}^n \alpha_i \langle h_i \rangle$

Such that $\mathcal{M}_Q = \{ \{ \langle h_i \rangle \} : \exists |\psi\rangle \in \mathcal{H} \text{ such that } \langle h_i \rangle = \langle \psi | h_i | \psi \rangle \quad \forall i \}$, where h_i are (possibly noncommuting) quantum observables and $\alpha_i \in \mathbb{R}$ are coefficients. In this sense, $\mathcal{M}_Q$ is just the possible combinations of expected energies $(\langle h_1 \rangle, \langle h_2 \rangle, \dots, \langle h_n \rangle)$ that are physically allowed by quantum mechanics.

These expected energies $\langle h_i \rangle$ are often called *moments*, and the set $\mathcal{M}_Q$ is the set of physically realizable quantum moments. For many-body systems, where $\dim(\mathcal{H})$ grows exponentially with system size, characterizing $\mathcal{M}_Q$ is generically extremely challenging.

Relaxation methods approach this difficulty by considering *supersets* $\mathcal{M}$ of $\mathcal{M}_Q$ that are easier to characterize. This superset contain all the physically possible energies but also other non physical energies, i.e energies that can't be written as $\langle h_i \rangle = \langle \psi | h_i | \psi \rangle \quad \forall i$ for a certain $|\psi\rangle$.

Optimizing over these relaxed sets provides *rigorous lower bounds* to the ground-state energy, complementing variational upper bounds and enabling systematic approximation schemes through hierarchies of relaxations.

While SDP's are not the only way to achieve relaxations of quantum problems, but in the context of quantum systems, they naturally lend themselves to this task due to the fundamental positivity and normalization constraints imposed by quantum mechanics.

2.1.2 SDP relaxations

Relaxations of polynomial optimization problems based on semi-definite programming (SDP) provide a systematic way to derive rigorous lower bounds on quantum ground-state energies and other observables. Because of their central role, we first outlined the main construction and implementation of these SDP relaxations in the context of ground-state problems.

These relaxations consider optimization problems of the form:

$$p_{\min} = \min_{\{|\psi\rangle, X\}} \langle \psi | p(X) | \psi \rangle \quad (2.1)$$

$$\begin{aligned} \text{subject to } & g_j(X) \succeq 0, \quad j = 1, \dots, m_1, \\ & \langle \psi | h_k(X) | \psi \rangle \geq 0, \quad k = 1, \dots, m_2. \end{aligned} \quad (2.2)$$

where p, g_j, h_k are polynomial functions of operator tuples $X = (X_1, \dots, X_n)$, and $\succeq 0$ denotes positive semidefiniteness.

One can readily see how the ground-state problem fits naturally within this polynomial optimization framework. Fermionic Hamiltonians, for example, are typically expressed as polynomials of creation and annihilation operators that satisfy canonical anticommutation relations, which themselves can be written as polynomial constraints. Hence, fermionic systems form a direct instance of the general polynomial constraints described in Eq.2.2. Similarly, for spin-1/2 systems, Hamiltonians are polynomial functions of Pauli matrices acting on each site. The Pauli matrices on a single site obey a well-known algebraic structure, while matrices on different sites commute. All these relations translate into polynomial constraints fitting into the relaxation formalism.

In practice we create a *quantum moments matrix* using all possible monomials constructed from elements of the tuple $X = (X_1, X_2, \dots)$ up to a certain degree. The maximum degree of these monomials define the *level* of our SDP. Generally increasing the level yields a better lower bound, but it also increases the computational cost. This can be used to create a hierarchy of optimization problems $p^1, p^2, \dots, p^\infty$ that should converge in principle to the exact ground energy E_{GS} . For smaller problems the exact solution can be obtained

at a finite rank.

An other way to improve the lower bound is to add more constraints, e.g symmetry relations, conservation of particles, commutation and anti-commutation relations, etc. However, it comes to no surprise that this also would increase the cost.

It may be interesting to note that not all constraints contribute equally to improving the lower bound, ergo a choice should be made to use the best ones, which is still an active area of research as there's no silver bullet to make the decision.

Building upon SDP relaxations, *quantum bootstrapping* has emerged as a powerful approach to estimate properties of complex quantum systems using limited data or smaller, more tractable subsystems [5, 18]. By exploiting symmetries, physical constraints, and prior information, quantum bootstrapping extrapolates from simpler models to infer the behavior of larger systems, thereby significantly reducing computational complexity. It is widely applied in quantum many-body physics and quantum computing to determine energies, correlation functions, and other observables without resorting to full-scale simulations.

Think of quantum bootstrapping like learning a big puzzle from a tiny piece: you solve a small, manageable part of a quantum system, extract the patterns or rules, and then use them to gradually reconstruct or approximate the behavior of the much larger, full system.

An other interesting aspect of applying SDP relaxations, is that we can find a rigorous lower bound on the expected value for any operator O that can be written as a polynomial functions of operator tuples $X = (X_1, \dots, X_n)$ of the set of operators. Once we found a lower bound E_A on the energy it can be added to the other constraints, along an upper bound E_R that can be obtained by variational methods. Thus we simply change the optimization problem to :

$$\begin{aligned} o_{\text{LB}} &= \min_{\{|\psi\rangle, X\}} \langle \psi | O(X) | \psi \rangle \\ \text{subject to } & g_j(X) \succeq 0, \quad j = 1, \dots, m_1, \\ & \langle \psi | h_k(X) | \psi \rangle \geq 0, \quad k = 1, \dots, m_2, \\ & \langle \psi | E_A - H(X) | \psi \rangle \geq 0, \\ & \langle \psi | H(X) - E_R | \psi \rangle \geq 0. \end{aligned} \tag{2.3}$$

By doing so, we are allowed to overcome the second drawback of variational methods mentioned earlier, and we can go beyond ground energy to other properties of the quantum system.

In the following section we try a small example for a small lattice of Graphene in the tight binding model.

2.2 Finding a lower bound value for graphene in the tight binding model

In the tight-binding model for graphene, we consider a honeycomb lattice composed of two sublattices, conventionally labeled A and B . Let $a_{\mathbf{R}}^\dagger$ and $b_{\mathbf{R}}^\dagger$ be the fermionic creation operators for an electron at position $\mathbf{R}$ on sublattices A and B , respectively. The second quantized Hamiltonian in real space for nearest-neighbor hopping reads:

$$H = -t \sum_{\mathbf{R}, i} \left(a_{\mathbf{R}}^\dagger b_{\mathbf{R}+\delta_i} + \text{h.c.} \right), \tag{2.4}$$

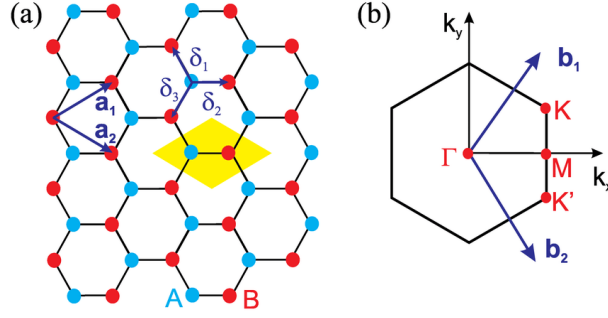


Figure 2.1: Caption

where t is the hopping parameter, δ_i ($i = 1, 2, 3$) are the vectors connecting each A site to its three neighboring B sites, and “h.c.” denotes the Hermitian conjugate.

To move to the reciprocal lattice (momentum space), we define the Fourier transforms:

$$a_{\mathbf{k}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{-i\mathbf{k} \cdot \mathbf{R}} a_{\mathbf{R}}, \quad b_{\mathbf{k}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{-i\mathbf{k} \cdot \mathbf{R}} b_{\mathbf{R}},$$

where N is the number of unit cells and $\mathbf{k}$ lies in the first Brillouin zone.

Substituting into the Hamiltonian, we obtain the momentum-space (reciprocal space) Hamiltonian:

$$H = \sum_{\mathbf{k}} \left[\phi(\mathbf{k}) a_{\mathbf{k}}^{\dagger} b_{\mathbf{k}} + \phi^*(\mathbf{k}) b_{\mathbf{k}}^{\dagger} a_{\mathbf{k}} \right],$$

where the structure factor $\phi(\mathbf{k}) = -t \sum_{i=1}^3 e^{i\mathbf{k} \cdot \delta_i}$ encodes the geometry of the lattice and interference effects due to hopping.

Since the operators $a_{\mathbf{k}}$ and $b_{\mathbf{k}}$ are fermionic, they obey the canonical anticommutation relations:

$$\{a_{\mathbf{k}}, a_{\mathbf{k}'}^{\dagger}\} = \delta_{\mathbf{k}, \mathbf{k}'}, \quad \{b_{\mathbf{k}}, b_{\mathbf{k}'}^{\dagger}\} = \delta_{\mathbf{k}, \mathbf{k}'}, \quad \text{others zero.}$$

This formalism allows for the diagonalization of the Hamiltonian using a suitable basis rotation (e.g., via a Bogoliubov transformation), leading to the well-known band structure of graphene featuring Dirac cones at the K and K' points. Applying

2.2.1 Analytical Calculation of the Ground-State Energy

In the nearest-neighbor tight-binding model of graphene, the Hamiltonian in momentum space for spinless electrons is

$$\hat{H}_0 = \sum_{\mathbf{k}} \begin{pmatrix} \hat{a}_{\mathbf{k}}^{\dagger} & \hat{b}_{\mathbf{k}}^{\dagger} \end{pmatrix} \begin{pmatrix} 0 & -tf(\mathbf{k}) \\ -tf^*(\mathbf{k}) & 0 \end{pmatrix} \begin{pmatrix} \hat{a}_{\mathbf{k}} \\ \hat{b}_{\mathbf{k}} \end{pmatrix}, \quad (2.5)$$

where $\hat{a}_{\mathbf{k}}$ and $\hat{b}_{\mathbf{k}}$ are annihilation operators for sublattices A and B , respectively, and

$$f(\mathbf{k}) = 1 + e^{i\mathbf{k} \cdot \mathbf{a}_1} + e^{i\mathbf{k} \cdot \mathbf{a}_2}, \quad (2.6)$$

with $\mathbf{a}_1$ and $\mathbf{a}_2$ the lattice vectors of the honeycomb lattice. Diagonalizing the 2×2 Hamiltonian for each $\mathbf{k}$ yields two energy bands:

$$E_{\pm}(\mathbf{k}) = \pm t |f(\mathbf{k})|. \quad (2.7)$$

At half-filling, the valence band ($E_-(\mathbf{k})$) is fully occupied and the conduction band ($E_+(\mathbf{k})$) is empty. The ground-state energy for the spinless system is therefore

$$E_0^{\text{spinless}} = \sum_{\mathbf{k}} E_-(\mathbf{k}) = - \sum_{\mathbf{k}} t |f(\mathbf{k})|. \quad (2.8)$$

In the thermodynamic limit, the sum over $\mathbf{k}$ can be replaced by an integral over the first Brillouin zone (BZ):

$$\frac{E_0^{\text{spinless}}}{N} = -t \frac{1}{\Omega_{\text{BZ}}} \int_{\text{BZ}} |f(\mathbf{k})| d^2k, \quad (2.9)$$

where N is the number of unit cells and Ω_{BZ} is the area of the BZ. This integral does not have a closed-form solution, but its numerical evaluation gives

$$\frac{E_0^{\text{spinless}}}{N} \approx -0.8075 t \quad (\text{per unit cell, spinless}). \quad (2.10)$$

Including electron spin doubles the number of occupied states, so the ground-state energy for spinful electrons is

$$E_0^{\text{spinful}} = 2 E_0^{\text{spinless}} \approx -1.615 t \quad \text{per unit cell}. \quad (2.11)$$

This analytical result provides a reference value for numerical simulations and can be used to benchmark calculations such as SDP relaxations for small graphene lattices.

2.2.2 Spin and Periodic Boundary Conditions in Tight-Binding Graphene

In the nearest-neighbor tight-binding model of graphene, the Hamiltonian for a spinless electron system is

$$\hat{H}_0 = -t \sum_{\langle i,j \rangle} \left(a_i^\dagger b_j + b_j^\dagger a_i \right), \quad (2.12)$$

where $\langle i, j \rangle$ denotes nearest-neighbor pairs and t is the hopping energy. In a finite lattice with periodic boundary conditions (PBC), each site at the edge of the lattice is connected to its neighbors across the boundary, effectively maintaining the translational symmetry of an infinite lattice. The ground-state energy of the spinless system at half-filling is given by

$$E_0^{\text{spinless}} = \sum_{n \in \text{occupied}} E_n, \quad (2.13)$$

where E_n are the eigenvalues of $\hat{H}_0$.

When spin is included, each lattice site can host two electrons (spin $\uparrow$ and $\downarrow$), and the Hamiltonian becomes

$$\hat{H} = \hat{H}_0 \otimes \mathbf{1}_{\text{spin}}. \quad (2.14)$$

Since the Hamiltonian contains no spin-dependent interactions, each single-particle energy level is doubly degenerate. Consequently, the ground-state energy for the spinful system is

$$E_0^{\text{spinful}} = 2 E_0^{\text{spinless}}. \quad (2.15)$$

This observation has important computational implications. When performing numerical methods such as semidefinite programming (SDP) relaxations, explicitly including spin operators would double the number of fermionic operators, increasing the size of the SDP matrices and memory requirements. By computing the spinless ground-state energy and multiplying by a factor of 2, one can obtain the spinful result while significantly reducing computational complexity. Moreover, periodic boundary conditions ensure that even a small lattice can mimic the translational invariance of the infinite graphene sheet, improving the accuracy of the finite-lattice approximation.

2.2.3 SDP for Graphene under the Tight binding model

The application of SDP of Graphene requires as explained earlier an expression of the energy as a polynomial of some operators, i.e the mean value of the Hamiltonian, and additional constraints under which we do the minimization.

Since a Hamiltonian in the reciprocal lattice is not physically significant unless we have a big number of sites N , and the number of sites N will increase the number of variables in our SDP real quick and becomes unfeasible (actually once implemented the problem is found to be unbounded even for very small N), we use the Hamiltonian in the real lattice given in Eq. 2.4.

We use the Hamiltonian expressed over lattice sites, and to keep the number of variables to a minimum we limit ourselves to 4 unit cells, giving 8 operators, with Born-Von Karmen boundary conditions to simulate a real lattice, i.e what goes out from one side must enter from the opposite side.

As for the constraints we use the fermionic Anti-commutation relations for which will serve for substitutions, and of course the Quantum moment matrix given by $M = (\langle v^\dagger u \rangle)_{v,u \in \Gamma}$ with Γ the set of monomials that we chose to work with.

Our SDP problem is then written as follows :

$$\begin{aligned} \text{minimize } E = \langle H \rangle &= \left\langle \sum_{1 \leq i, j \leq 3} a_i^\dagger b_j \right\rangle \\ \text{subject to } * M &\succeq 0, \\ * \{a_{\mathbf{k}}, a_{\mathbf{k}'}^\dagger\} &= \delta_{\mathbf{k}, \mathbf{k}'}, \quad \{b_{\mathbf{k}}, b_{\mathbf{k}'}^\dagger\} = \delta_{\mathbf{k}, \mathbf{k}'}, \quad \text{others zero.} \end{aligned} \quad (2.16)$$

To solve the problem we write a python code, using a library called `ncpol2sdpa` [19]. Given arrays of operators, a hamiltonian, and a dictionary containing the constraints and the desired level, This library translate it for an SDP solver that is optimised to solve the optimization problem, for the work we went with solver `SDPA`.

Following the analytical treatment in 2.2.1, and using a hopping term $t = 2.8eV$, we expect to get a lower bound that gets closer to $E_{GS} = -4.522eV$. Also to observe the effect of increasing the level and adding new constraints we groupe the result for several executions in the table 2.1. `Extra anticommutation=true` means that we used anticommutation relations between operators a_i and b_j and not only between operators of the same lattice, otherwise it set to `false`. The column `Solver status` is the outcome of the attempt of the solver to find a solution.

The files for executing each step of this workflow are provided in the accompanying GitHub repository.

Table 2.1: Ground-state energy results for Graphene at different calculation levels and anticommutation settings, including solver status.

Calculation	Level	Anticommutation	Solver Status	Energy (eV)
Run 1	1	False	Optimal	Ground State: -89.6000 Per Unit Cell: -22.4000
Run 2	1	True	Optimal	Ground State: -78.4000 Per Unit Cell: -19.6000
Run 3	2	False	Primal-dual Feasible	Ground State: -54.0784 Per Unit Cell: -13.5196
Run 4	2	True	Optimal	Ground State: -27.0392 Per Unit Cell: -6.7598

Chapter 3

Applications and Results for Silicon

This chapter details the practical implementation of the first-principles workflow for silicon, outlines the computational challenges encountered, and discusses alternative approaches for future work. The goal was to construct a first-principles Hamiltonian for silicon using the established DFT-Wannier90-Respack pipeline, then apply semidefinite programming (SDP) methods to obtain certified bounds on ground-state properties beyond energy.

All files used in this work shall be put in a Github repository : <https://github.com/Amddah-Mohamed/Si-Hamiltonian-with-QE-Wannier90-for-SDP>

3.1 Target Workflow for Silicon Hamiltonian Construction

The planned computational pipeline followed the methodology outlined in Chapter 1 and mirrored the Phascraft. The workflow consisted of three main stages:

3.1.1 Stage 1: Density Functional Theory with Quantum ESPRESSO

The initial stage involved performing first-principles DFT calculations using Quantum ESPRESSO (QE). For silicon, which crystallizes in a diamond structure (space group $Fd\bar{3}m$), we used:

- A primitive unit cell with 2 silicon atoms
- Optimized norm-conserving Vanderbilt (ONCV) pseudopotentials
- Plane-wave energy cutoff of 60 Ry
- $8 \times 8 \times 8$ $\mathbf{k}$ -point mesh for Brillouin zone sampling
- Perdew-Burke-Ernzerhof (PBE) exchange-correlation functional

The QE calculation proceeds through two self-consistent loops (Figure 1.2): the inner loop converges the electronic ground state, while the outer loop optimizes the atomic positions. The primary outputs of this stage are:

- `silicon.scf.out`: Self-consistent total energy, forces, and stress tensor
- `silicon.nscf.out`: Bloch wavefunctions $\psi_{n\mathbf{k}}(\mathbf{r})$ on a dense $\mathbf{k}$ -point grid
- Electronic band structure (Figure 3.1a), showing the characteristic indirect gap between Γ and X points
- Density of states (DOS), revealing the sp^3 hybridization around the Fermi level

These outputs provide the fundamental electronic structure information necessary for constructing localized orbitals.

3.1.2 Stage 2: Wannierization with Wannier90

The Bloch wavefunctions from QE were processed using Wannier90 [9,16] to generate Maximally Localized Wannier Functions (MLWFs). For silicon, we selected the four valence bands (corresponding to the sp^3 hybrid orbitals) as the active space. The Wannier90 calculation:

- Initialized with atomic sp^3 trial projections
- Minimized the spread functional Ω (Eq. 1.19) through iterative unitary rotations
- Generated MLWFs centered on Si-Si bonds (Figure 3.1b)

Key outputs included:

- `silicon_hr.dat`: Real-space hopping matrix $T(\mathbf{R})_{mn}$ 1.16
- `silicon.wout`: MLWF centers, spreads, and convergence information
- Visualization of MLWFs showing their chemical bonding character

The hopping matrix `silicon_hr.dat` defines the single-particle Hamiltonian $\hat{H}_0$ for the tight-binding model:

$$\hat{H}_0 = \sum_{ij,\sigma} t_{ij} c_{i\sigma}^\dagger c_{j\sigma}$$

This compact representation captures the essential electronic structure with dramatically reduced computational cost compared to full DFT.

3.1.3 Stage 3: Electron-Electron Interactions with Respack

The final stage involved calculating the Coulomb interaction tensor using RESPACK [?], which implements the constrained random-phase approximation (cRPA) method. The intended calculation would have:

- Used the MLWFs from Wannier90 as the orbital basis
- Computed the screened Coulomb interaction $W(\mathbf{r}, \mathbf{r}')$
- Generated the full interaction tensor V_{ijkl} (Eq. 1.18)

The complete many-body Hamiltonian would then be:

$$\hat{H} = \hat{H}_0 + \frac{1}{2} \sum_{ijkl,\sigma\sigma'} V_{ijkl} c_{i\sigma}^\dagger c_{j\sigma'}^\dagger c_{k\sigma'} c_{l\sigma}$$

3.2 Computational Challenges and Limitations

Despite careful planning and implementation of the QE and Wannier90 stages, significant compatibility issues arose when attempting to interface with Respack. The primary challenges were:

- **Version incompatibility:** The specific Respack version required for silicon calculations demanded an older version of Wannier90 than used in this work
- **File format discrepancies:** Subtle differences in the Wannier90 output format prevented seamless data transfer to Respack
- **Numerical instability:** Preliminary attempts to manually bridge the compatibility gap resulted in numerically unstable Coulomb tensor elements

These technical hurdles prevented the calculation of the full Coulomb interaction tensor within the project timeframe. Consequently, we could not construct the complete many-body Hamiltonian required for SDP simulations targeting electron correlation effects.

3.3 Alternative Approach: Monte Carlo Integration

Given the compatibility issues with Respack, an alternative pathway for future work involves direct Monte Carlo integration to compute the Coulomb tensor elements. The general formula for the four-index integral is:

$$V_{ijkl} = \frac{1}{2} \int d\mathbf{r} d\mathbf{r}' w_i^*(\mathbf{r}) w_j^*(\mathbf{r}') V(|\mathbf{r} - \mathbf{r}'|) w_k(\mathbf{r}') w_l(\mathbf{r})$$

Monte Carlo integration estimates this high-dimensional integral by sampling:

$$V_{ijkl} \approx \frac{1}{N} \sum_{n=1}^N w_i^*(\mathbf{r}_n) w_j^*(\mathbf{r}'_n) V(|\mathbf{r}_n - \mathbf{r}'_n|) w_k(\mathbf{r}'_n) w_l(\mathbf{r}_n)$$

where $(\mathbf{r}_n, \mathbf{r}'_n)$ are randomly sampled points in the simulation cell. This approach offers several advantages:

- **Implementation independence:** Avoids software compatibility issues by working directly with MLWF numerical data
- **Progressive accuracy:** Results improve systematically with increased sampling
- **Selective calculation:** Can focus computational effort on dominant interaction terms identified through the Cauchy-Schwarz inequality (Eq. 1.8)

The MLWFs from Wannier90 are ideally suited for this approach as they are real-valued and exponentially localized, allowing for efficient importance sampling around their centers.

3.4 Utilization of Obtained Results

Despite the inability to complete the full workflow, the successful execution of the QE and Wannier90 stages provides valuable resources for further investigation:

3.4.1 Band Structure and Density of States

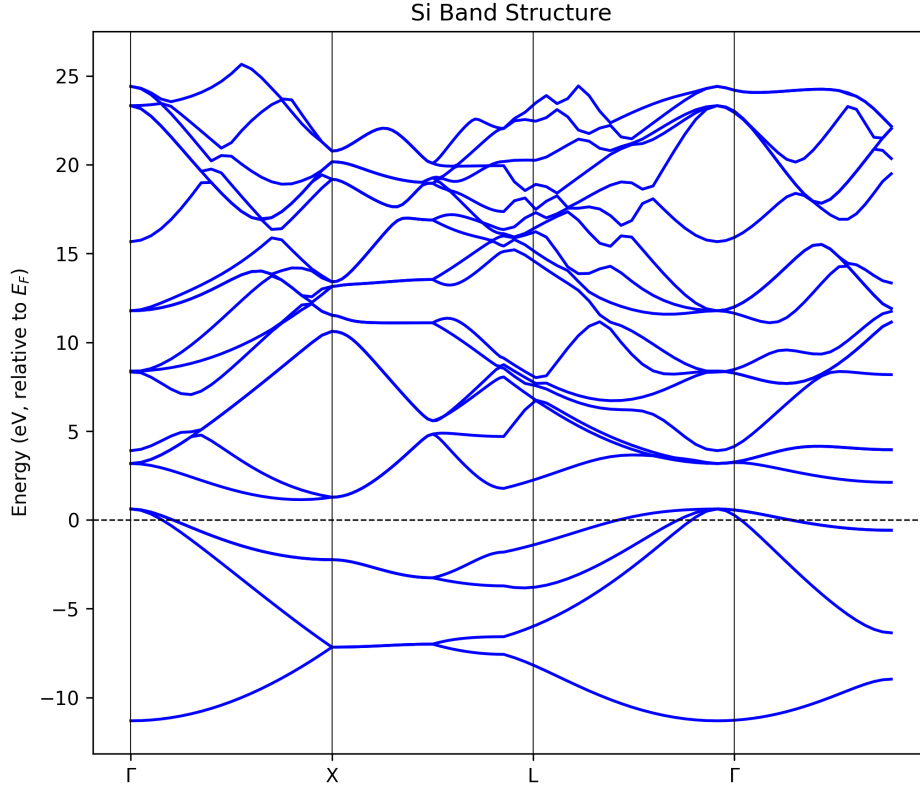


Figure 3.1: Band structure of silicon.

The MLWF-based Hamiltonian (`silicon_hr.dat`) accurately reproduces the DFT band structure when Fourier transformed back to momentum space. This enables efficient calculation of:

- Electronic band structure along high-symmetry paths
- Density of states (DOS) and projected DOS (PDOS)
- Effective masses at band extrema
- Fermi surfaces for doped systems

3.4.2 Large-Scale Simulations

The localized nature of MLWFs enables efficient simulations of large systems containing thousands of atoms, including:

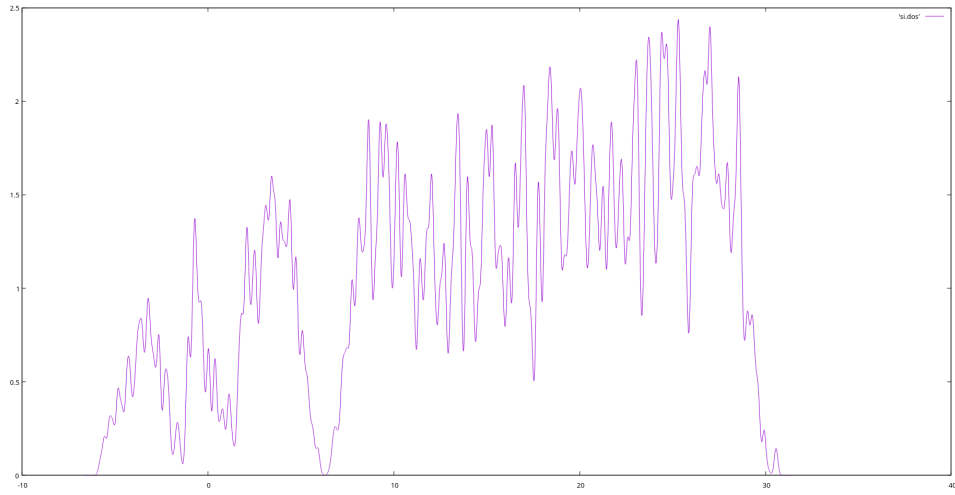


Figure 3.2: Density of states of silicon.

- Defect calculations (vacancies, impurities)
- Surface and interface studies
- Disorder effects through statistical sampling

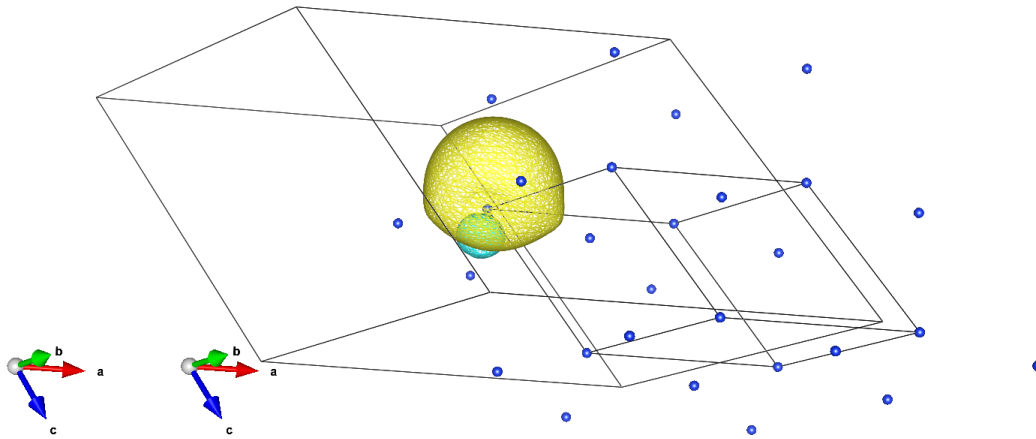


Figure 3.3: Realspace Wannier function. The 1st Wannier function for sp^3 orbital in Si, with a spread of $2.7 \text{ \AA}^2$. The figure is made using the visualization software VESTA (<https://jp-minerals.org/vesta/jp/>)

Conclusion

Although technical compatibility issues prevented the full implementation of the SDP workflow for silicon, this project successfully established a reliable Quantum ESPRESSO–Wannier90 pipeline and demonstrated the construction of maximally localized Wannier functions. The resulting MLWF representation provides a compact and physically transparent description of silicon’s electronic structure, which can already be applied to band interpolation, model Hamiltonian construction, and analysis of bonding characteristics. One can also use it to find improved band structure with interpolations of MLWFs.

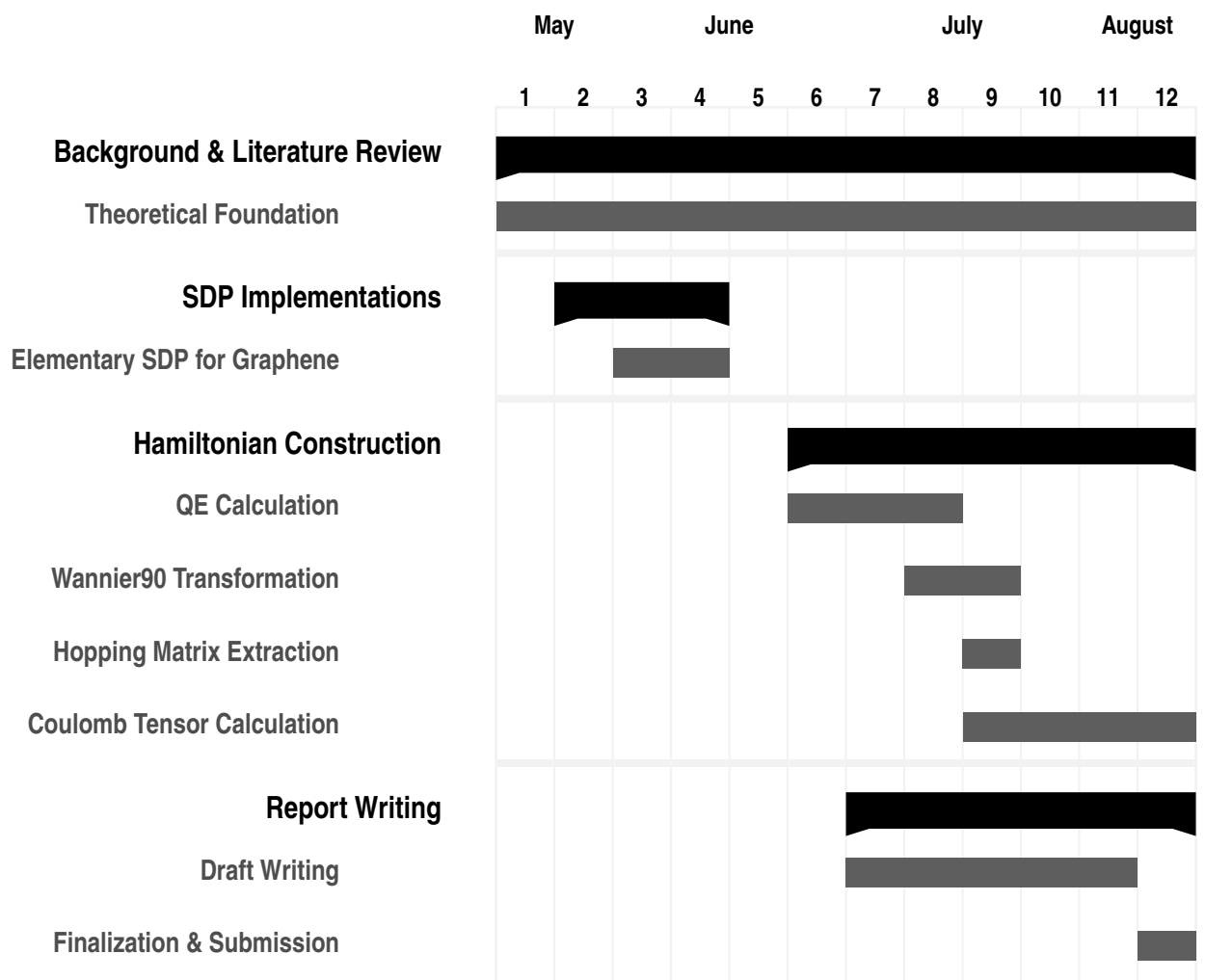
The internship also provided a great opportunity to learn using important softwares for simulating quantum systems and also to discover a deeper understanding of the state of the art in computational physics.

For advancing toward the original objective of rigorously certified ground-state properties, two promising directions remain: (i) resolving the RESPACK integration issues to enable the direct generation of interaction Hamiltonians, or (ii) implementing alternative numerical approaches such as Monte Carlo integration. Pursuing either pathway would open the door to applying semidefinite programming (SDP) techniques not only for ground-state energy bounds but also for extracting higher-order properties such as charge density correlations, dielectric response, and excitation gaps. In this way, the groundwork done here provides both a valuable toolset and a clear roadmap for future work at the intersection of electronic-structure methods and convex optimization.

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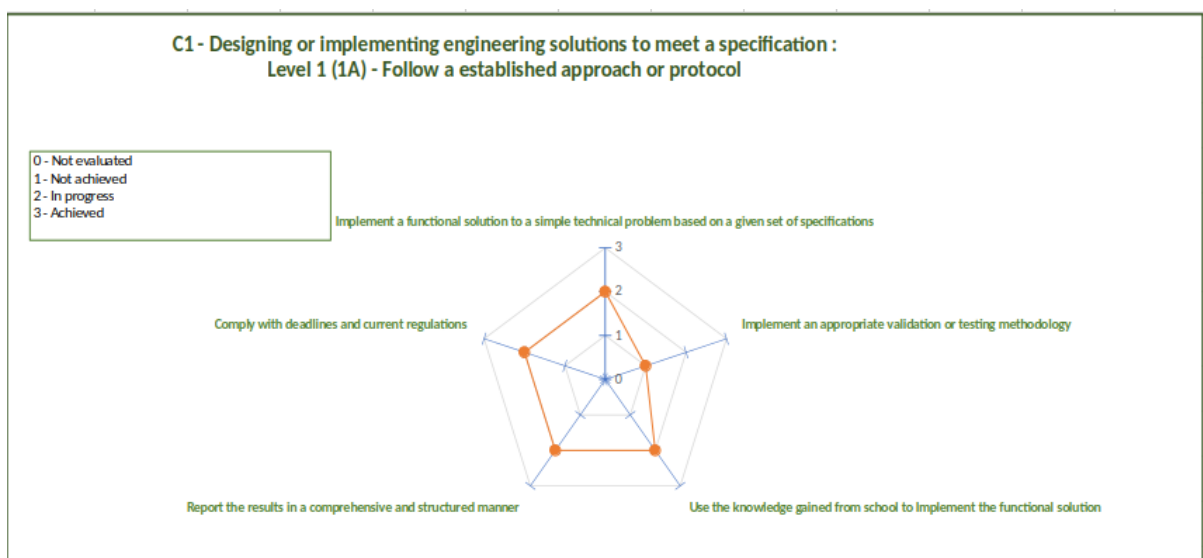
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C1 - Designing or implementing engineering solutions to meet a specification

C1 - Designing or implementing engineering solutions to meet a specification	C1 - Designing or implementing engineering solutions to meet a specification (Level 1)	
	Intended Learning Outcomes (ILO) - Level 1 (Follow a established approach or protocol)	Score
	Implement a functional solution to a simple technical problem based on a given set of specifications	2
	Implement an appropriate validation or testing methodology	1
	Use the knowledge gained from school to Implement the functional solution	2
	Report the results in a comprehensive and structured manner	2
	Comply with deadlines and current regulations	2
	C1 - Designing or implementing engineering solutions to meet a specification (Level 2)	
	Intended Learning Outcomes (ILO) - Level 2 (Propose an approach or a protocol)	Score
	Leverage one's knowledge and external resources	2
	Design and implement a functional solution including resources and timelines	1
	Test the solution and conduct a critical analysis	2
	Document the work (solution, methods, tests, etc.) for future reuse by others	3



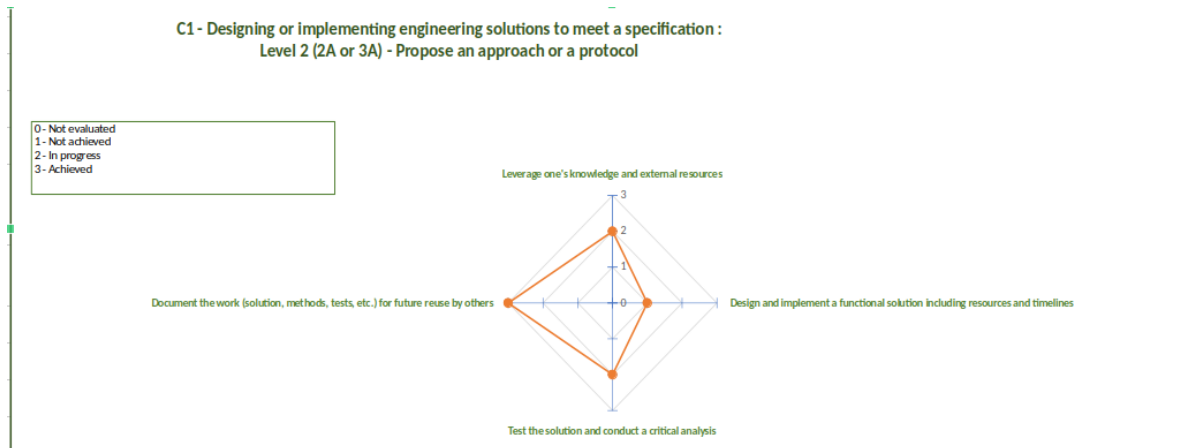


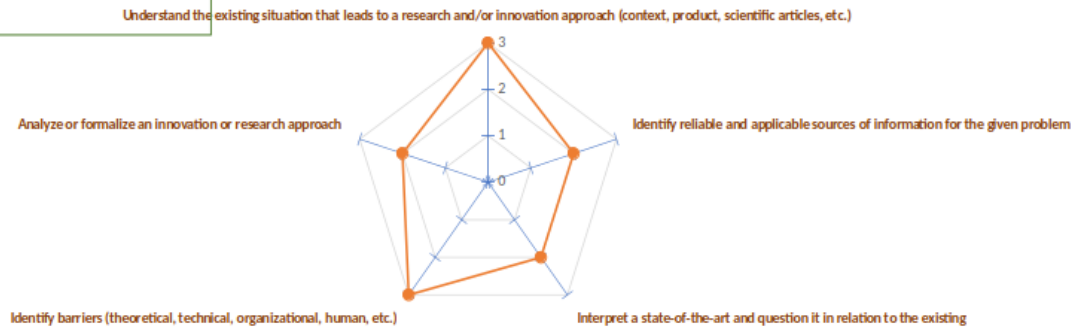
Figure 3.4: Caption

C2 - Implementing a fundamental research approach or an applied research approach for innovation purposes

C2 - Implementing a fundamental research approach or an applied research approach for innovation purposes	C2 - Implementing a fundamental research approach or an applied research approach for innovation purposes (Level 1)	
	Intended Learning Outcomes (ILO) - Level 1 (1A or 2A) : Analyze and utilize an innovation or research approach	Score
	Understand the existing situation that leads to a research and/or innovation approach (context, product, scientific articles, etc.)	3
	Identify reliable and applicable sources of information for the given problem	2
	Interpret a state-of-the-art and question it in relation to the existing	2
	Identify barriers (theoretical, technical, organizational, human, etc.)	3
	Analyze or formalize an innovation or research approach	2
	C2 - Implementing a fundamental research approach or an applied research approach for innovation purposes (Level 2)	
	Intended Learning Outcomes (ILO) - Level 2 (3A) : Carry out a research or innovation process	Score
	Suggest and develop new approaches (technical or theoretical, implementation paths, optimizations, etc.) by incorporating new ideas	1
	Evaluate the feasibility of the envisaged new approaches using logical arguments to persuade	2
	Demonstrate autonomy by implementing a process of goal setting and decision-making	3
	Present in a reasoned and clear manner any element related to the new approach (state of the art, work done) with perspective and critical thinking	2

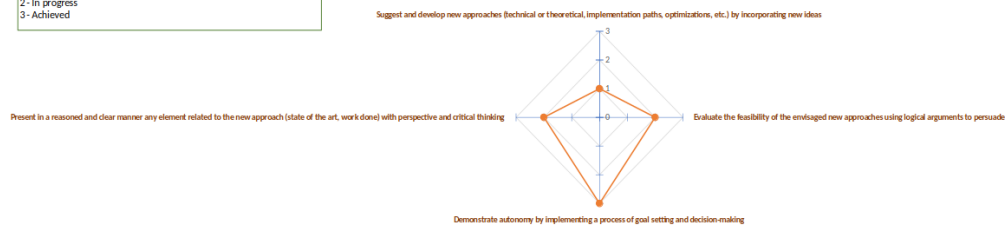
C2 - Implementing a fundamental research approach or an applied research approach for innovation purposes :
Level 1 (1A or 2A) - Analyze and utilize an innovation or research approach

0 - Not evaluated
 1 - Not achieved
 2 - In progress
 3 - Achieved



C2 - Implementing a fundamental research approach or an applied research approach for innovation purposes :
Level 2 (3A) - Carry out a research or innovation process

0 - Not evaluated
 1 - Not achieved
 2 - In progress
 3 - Achieved

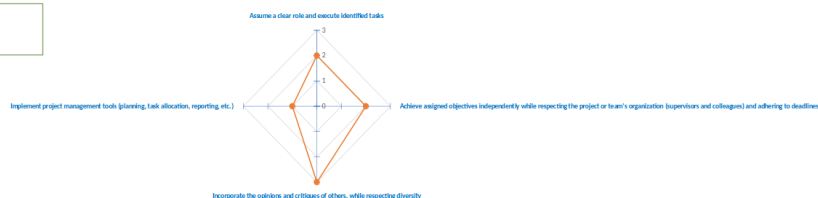


C3 - Collaborate within a team and in project mode

C3 - Collaborate within a team and in project mode	C3 - Collaborate within a team and in project mode (Level 1)	
	Intended Learning Outcomes (ILO) - Level 1 (1A) : Participate in a team	Score
	Assume a clear role and execute identified tasks	2
	Achieve assigned objectives independently while respecting the project or team's organization (supervisors and colleagues) and adhering to deadlines	2
	Incorporate the opinions and critiques of others, while respecting diversity	3
	Implement project management tools (planning, task allocation, reporting, etc.)	1
	C3 - Collaborate within a team and in project mode (Level 2)	
	Intended Learning Outcomes (ILO) - Level 2 (2A or 3A) : Lead/facilitate a project	Score
	Structure a project (breaking it down into subtasks, planning, mobilizing the necessary human, technical, or material resources)	2
	Perform complex actions while interacting with external professionals in an environment conducive to autonomy, requiring thoughtful task allocation and initiative and organizational skills	2
	Monitor the progress of the project and adjust the work accordingly (crisis management, conflict resolution, etc.). Make the necessary decisions to achieve the objectives.	1
	Communicate the results of the overall project to decision-makers	3

C3 - Collaborate within a team and in project mode :
Level 1 (1A) - Participate in a team

0- Not evaluated
1- Not achieved
2- In progress
3- Achieved



C3 - Collaborate within a team and in project mode :
Level 2 (2A or 3A) - Lead/facilitate a project

0- Not evaluated
1- Not achieved
2- In progress
3- Achieved

